

Skills Summary

Random Samples and Simple Probability Models

Use this reference while completing your worksheet or when studying.

A Simple Probability Model

Rule: when every outcome is equally likely,

$$P(\text{event}) = \frac{\text{favourable outcomes}}{\text{total outcomes}}$$

Every probability lands between 0 and 1. Reduce the fraction. “Not the event” has probability $1 - P(\text{event})$.

Example: A tray holds 15 identical cards and 6 of them are yellow. One card is drawn at random. Find the probability it is yellow.

Step 1. The cards are identical, so every card is equally likely — count the favourable ones over the total.

Step 2. Six of the fifteen cards are yellow, so the probability is $\frac{6}{15}$, which reduces to **2/5**.

Try it: A box holds 24 pens and 9 of them are red. Find the probability of drawing a red pen.

check: 3/8

Scaling a Random Sample up to a Population

Rule: a random sample estimates the whole population by keeping the same fraction:

$$\text{estimate} = \frac{\text{count in the sample}}{\text{size of the sample}} \times \text{size of the population}$$

Write the sample fraction first, then multiply. Always state the units of the answer.

Example: In a random sample of 40 books from a library of 500, 12 were poetry. Estimate how many poetry books the library holds.

Step 1. The sample was random, so the fraction of poetry in it is a fair estimate of the fraction in the library.

Step 2. The sample fraction is $\frac{12}{40} = 0.3$, so the estimate is $0.3 \times 500 = \mathbf{150}$ books.

Try it: In a random sample of 35 apples from a crop of 400, 7 were bruised. Estimate the number of bruised apples in the crop.

check: 08

Judging Whether a Sample Can Be Trusted

Rule: a sample is **random** when every member of the population has the same chance of being chosen. If it does not, the estimate describes the group you asked, not the population — and no amount of correct arithmetic repairs that.

Warning signs: the group was chosen because of the very thing being measured, or people chose themselves by turning up, replying, or volunteering.

Example: A town of 450 residents is sampled by numbering every resident and drawing 30 numbers at random. What chance does one named resident have of being chosen?

Step 1. Every resident is numbered, so every one is equally likely — that is what makes the sample random, and it also makes the chance easy to count.

Step 2. Thirty of the 450 residents are drawn, so a named resident has probability $\frac{30}{450} = 1/15$.

Try it: A school numbers all 500 students and draws 25 at random. What is the probability that one named student is chosen?

check: 1/20

(!) Watch out: surveying the basketball team about the school's favourite sport gives a perfectly correct calculation and a useless estimate. Check how the sample was chosen *before* you scale it up.